FAMILY NAME: ....
OTHER NAME(S): ....
STUDENT NUMBER: ....
SIGNATURE: ....

UNSW, SYDNEY

SCHOOL OF MATHEMATICS AND STATISTICS

SUPPLEMENTARY EXAMINATION

Term 1, 2023

MATH5975
Introduction to Stochastic Analysis

(1) TIME ALLOWED – 2 Hours

(2) TOTAL NUMBER OF QUESTIONS – 5

(3) ANSWER ALL QUESTIONS

(4) READING TIME – 10 Minutes

(5) THE QUESTIONS ARE **NOT** OF EQUAL VALUE

(6) TOTAL NUMBER OF MARKS – 50

(7) START EACH QUESTION ON A NEW PAGE.

(8) THIS PAPER MAY **NOT** BE RETAINED BY THE CANDIDATE

(9) **ONLY** CALCULATORS WITH AN AFFIXED "UNSW APPROVED" STICKER MAY BE USED

All answers must be written in ink. Except where they are expressly required pencils may only be used for drawing, sketching or graphical work.

1. [**8 marks**] Let $\Omega = \{1,2,3,4\}$ and we consider the σ-algebra given by the power set 2^Ω. Let $\mathbb{P}$ defined on 2^Ω be given by $\mathbb{P}(1) = \mathbb{P}(2) = \mathbb{P}(3) = \mathbb{P}(4) = \frac{1}{4}$.

i) Write down the σ-algebra generated by $A = \{1,2\}$ given by $\sigma(A)$.

ii) Consider the random variables X and $Y : \Omega \to \mathbb{R}$ given by $X(\omega) = \omega$ and $Y = X^2$.

a) Compute $\mathbb{E}[X]$ and $\mathbb{E}[Y]$.

b) Compute $\mathbb{E}[X|Y]$.

iii) Consider another random variable Z given by

$$Z = \begin{cases} 1 & X \geq 2 \\ -1 & X < 1, \end{cases}$$

and compute $\mathbb{E}[X|Z]$.

2. [**8 marks**] Let W be a one-dimensional Wiener process on the space $(\Omega, \mathcal{F}, \mathbb{P})$.

i) Let $X_t = \int_0^t W_u dW_u$ for $t \in [0,T]$. Show that X is a martingale (not just a local martingale) and compute it's quadratic variation $\langle X \rangle$.

ii) Show that X admits the following representation

$$X_t = \frac{1}{2}(W_t^2 - t).$$

iii) Let $X_t = \int_0^t \sin(u) dW_u$ for $t \in [0,T]$

a) Compute $\mathbb{E}(X_t)$ and $\mathrm{Cov}(X_t, X_s)$ for $s,t > 0$.

b) Compute the quadratic variation of X, that is $\langle X \rangle_t$ for $t \in [0,T]$. *Hint:* $\cos(2x) = 1 - 2\sin^2(x)$.

3. [**10 marks**] Let $W_t = (W_t(1), W_t(2))$ be a standard two dimensional Brownian motion and let R_t be the distance the Brownian motion is away from the origin at time $t > 0$, that is $R_t^2 = W_t^2(1) + W_t^2(2)$. In the following, you may assume that $\mathbb{P}(R_t > 0, \forall t > 0) = 1$ and $1/R_t$ is well-defined and the Itô formula can be applied.

i) Compute the quadratic variation of R denoted by $\langle R \rangle$.

ii) By applying the Itô formula, show that $Y_t = -\log(R_t)$ is a local martingale for $t \in [1,T]$.

iii) By showing that $\mathbb{E}[Y_t]$ is not a constant and hence deduce that Y_t for $t \in [1,T]$ is not a martingale.

Hint: You may use the fact that χ_k^2 *distribution is the same as Gamma*$(\frac{k}{2}, 2)$.

4. **[14 marks]** Given a Brownian motion W, the Brownian bridge is given by

$$X_t := W_t - tW_1, \quad t \in [0,1];$$

i) Compute the expected value and the covariance function of X.

ii) Explain why $X = (X_t)_{t\in[0,1]}$ is not $\mathcal{F}_t^W$-measurable and show that the process X is adapted to the filtration $\mathbb{G} = (\mathcal{G}_t)_{t\geq 0}$ where $\mathcal{G}_t = \sigma(\mathcal{F}_t^W, W_1)$.

$$\mathbb{E}(W_t - W_s|\mathcal{G}_s) = \frac{t-s}{1-s}(W_1 - W_s).$$

Hint: Given a multivariate Gaussian random vector $(X,Y) \in \mathbb{R}^2$ *with mean vector* $(\mu_X, \mu_Y) \in \mathbb{R}^2$*, it can be shown that*

$$\mathbb{E}(X|Y=y) = \mu_X - \mathrm{Cov}(X,Y)[\mathrm{Var}(Y)]^{-1}(y - \mu_Y).$$

iii) Is $X = (X_t)_{t\in[0,1]}$ a $\mathbb{G}$-martingale?

5. **[10 marks]** The square root process X is defined as the solution of the stochastic differential equation

$$\begin{aligned} dX_t &= (a - bX_t)dt + \sqrt{X_t}dB_t, \quad t \in [0,\infty) \\ X_0 &= x \end{aligned} \tag{1}$$

where a, b and x are positive constants. In the following, you may assume that the solution process X is positive.

i) Show that the process $Y_t = X_t^{\frac{1}{2}}$ satisfies the equation

$$\begin{aligned} dY_t &= -\frac{1}{2}bY_t dt + \frac{1}{2}dB_t + \frac{1}{2}(a - \frac{1}{4})\frac{1}{Y_t}dt \\ Y_0 &= \sqrt{x} \end{aligned}$$

ii) Let $a = 1/4$, find the explicit solution of Y and hence X.

iii) Let $a = 1/4$ and find $\mathbb{E}(Y_t)$ and $\lim_{t\to\infty}\mathbb{E}(Y_t)$.

END OF EXAMINATION

Table of some common distributions

X	mass/density function	domain	mean	variance	mgf $m_X(u) = E(e^{uX})$
Bernoulli	$P(X=1) = p$ $P(X=0) = q = 1-p$	$\{0, 1\}$	p	pq	$q + pe^u$
Binomial $Bin(n, p)$	$\binom{n}{k} p^k (1-p)^{n-k}$	$k = 0, 1, \ldots, n$	np	$np(1-p)$	$(1 - p + pe^u)^n$
Geometric	$p(1-p)^{k-1}$	$k = 1, 2, \ldots$	$\frac{1}{p}$	$\frac{(1-p)}{p^2}$	$\frac{p}{e^{-u} - 1 + p}$
Poisson	$e^{-\lambda} \lambda^k / k!$	$k = 0, 1, \ldots$	λ	λ	$\exp\{\lambda(e^u - 1)\}$
Uniform	$(b-a)^{-1}$	$x \in (a, b)$	$\frac{1}{2}(a+b)$	$\frac{1}{12}(b-a)^2$	$\frac{e^{bu} - e^{au}}{u(b-a)}$
Exponential	$\frac{1}{\beta} e^{-\frac{1}{\beta} x}$	$x \in (0, \infty)$	β	β^2	$\frac{1}{1 - \beta u}$
Normal $N(\mu, \sigma^2)$	$\frac{1}{\sqrt{2\pi\,\sigma^2}} \exp\left\{ -\frac{(x-\mu)^2}{2\sigma^2} \right\}$	$x \in \{-\infty, \infty\}$	μ	σ^2	$e^{\mu u + \frac{1}{2}\sigma^2 u^2}$
Gamma (α, β)	$\frac{e^{-x/\beta} x^{\alpha-1}}{\Gamma(\alpha)\beta^\alpha}$	$x \in (0, \infty)$	$\alpha\beta$	$\alpha\beta^2$	$\left(\frac{1}{1-\beta u}\right)^\alpha$
Beta $\beta(\alpha, \beta)$	$\frac{\Gamma(\alpha+\beta)}{\Gamma(\alpha)\Gamma(\beta)} x^{\alpha-1}(1-x)^{\beta-1}$	$x \in [0, 1]$	$\frac{\alpha}{\alpha+\beta}$	$\frac{\alpha\beta}{(\alpha+\beta)^2(\alpha+\beta+1)}$	